

SUPERIOR GRAND TRAVERSE BAY, MI, USA

WMO: 998176

Lat: 47.179N

Lon: 88.241W

Elev: 627

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-2.8	N/A	N/A	N/A	N/A	N/A	N/A	25.1	28.6	23.0	28.1	6.8	320	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	82.8	N/A	79.3	N/A	75.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.6	280

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	(e)	(f)
24.6	20.3	17.7	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)
			DB	-14.7	88.6	8.5	2.4	-20.9	90.4	-25.8	91.8	-30.6	93.1	-36.8	94.8	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	16.5	16.4	24.7	36.1	45.1	55.0	63.8	63.5	56.5	45.0	34.2	23.5
	DBStd	18.53	10.31	10.31	10.63	6.53	6.65	7.26	6.04	5.78	7.23	7.28	8.70	9.26
	HDD50	4910	1037	940	785	421	186	29	0	0	21	192	476	823
	HDD65	9209	1502	1360	1250	868	618	311	96	100	271	621	924	1288
	CDD50	1315	0	0	0	3	33	178	429	417	215	37	3	0
	CDD65	138	0	0	0	0	0	10	60	52	15	1	0	0
	CDH74	1050	0	0	0	0	5	113	405	409	105	13	0	0
	CDH80	216	0	0	0	0	0	20	74	101	20	1	0	0
Wind	WSAvg	7.1	7.7	8.1	7.9	8.6	7.3	6.1	5.9	6.0	6.2	7.7	6.6	6.9
Precipitation	PrecAvg	26.7	1.6	1.2	1.2	2.0	2.8	2.9	2.8	2.2	3.3	2.9	2.0	1.7
	PrecMax	35.1	4.5	2.6	3.1	5.6	6.4	5.6	6.5	5.4	9.9	7.4	4.7	3.4
	PrecMin	19.1	0.0	0.4	0.0	0.6	1.3	0.6	1.1	0.4	1.2	1.0	0.6	0.3
	PrecStd	4.4	1.3	0.6	0.8	1.2	1.1	1.2	1.2	1.2	1.7	1.5	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	41.5	53.8	65.1	75.1	82.4	86.1	87.5	82.9	75.0	59.7	47.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	37.7	36.6	48.0	56.3	68.4	77.4	82.4	83.2	76.6	65.8	52.8	41.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	35.1	33.8	43.8	49.6	61.7	72.1	79.2	78.7	72.2	60.2	49.9	38.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	32.4	31.6	39.3	45.5	56.9	66.7	75.7	74.9	68.8	56.4	47.3	36.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	12.3	15.7	18.8	14.4	15.6	16.3	18.2	18.5	18.5	14.0	11.9	11.4
		MCDBR	14.0	18.1	23.4	22.0	25.5	25.4	23.0	25.1	24.3	23.1	17.6	12.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.266	0.286	0.310	0.345	0.376	0.383	0.394	0.375	0.365	0.329	0.299	0.267	
	taud	2.441	2.364	2.373	2.362	2.372	2.370	2.391	2.442	2.455	2.521	2.522	2.476	
	Ebn at Noon	266	281	290	288	282	279	274	276	268	263	251	253	
	Edh at Noon	21	28	32	36	37	37	36	33	30	24	20	18	
All-Sky Solar Radiation	RadAvg	318	611	1080	1467	1752	1911	1945	1661	1208	713	360	234	
	RadStd	30	73	109	163	124	133	90	110	68	86	53	28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page